Proxy Rotation

Description	Rotation to avoid a Proxy-BlackList

The proxy rotation is so important why if an IP address is used for a nefarious activity, it can be blacklisted in a DNS-based anti-spam database.
An activity such as spamming could get an IP address entered into such a database.

In fact changing proxies is a common practice in web scraping for several key reasons:

1. `Avoiding IP Blocking:`
    - By regularly changing your IP address through different proxies, you can mitigate the risk of being blocked by a website due to an excessive number of requests from a single IP address. This practice helps to maintain a stable connection to the website without interruptions.
2. `Anonymity:`
    - Changing proxies allows you to hide your identity and actual location, providing a certain level of anonymity during web scraping. This is useful when you want to protect your identity or when a website may restrict access based on geographical location.
3. `Overcoming Geographical Restrictions:`
    - By changing the proxy, you can simulate presence in different geographic regions, enabling access to online content that may be restricted to specific locations.
4. `Load Distribution:`
    - If conducting web scraping on a large scale, changing the proxy helps distribute the load among different IP addresses. This helps avoid overloading a single website server and reduces the likelihood of being blocked.
5. `Maintaining Separate Sessions:`

- In situations where it is necessary to maintain separate sessions (e.g., during authentication through a login process), changing the proxy can simulate the behavior of different users, allowing the maintenance of distinct sessions.

6. `Avoiding Bot Detection:`
    - Changing the proxy reduces the likelihood of being detected as a bot. Some websites employ automated bot detection techniques, and changing the IP address can help bypass such detection systems.

Pseudo Code

```
proxy_list = [
    'http://proxy1.example.com:port',
    'http://proxy2.example.com:port',
    # add other proxy necessary
]

# create an infinity cicle for the proxy rotation
proxy_pool = cycle(proxy_list)

# get a random proxy from an infinity cycle
proxy = get_random(proxy_pool)

# set the random proxy for the request
proxies = {'http': proxy, 'https': proxy}
```